

Pengfei Shen

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Google Scholar

Research Interests

Computational Imaging, Differentiable Rendering, Realistic Rendering

Education

- Ph.D. Electrical and Computer Engineering, The University of Hong Kong, 2024–present
Advised by Prof. Yifan (Evan) Peng
- M.S. Applied Mathematics, University of Science and Technology of China, 2021–2024
Advised by Prof. Ligang Liu
- B.S. Physics, University of Science and Technology of China, 2017–2021
Dual B.E. in Computer Science
Thesis: *An Improved Rendering Method: Consideration of Optical Polarization Effect in Scattering Process*

Publications

- Li Liao[†], **Pengfei Shen**[†], and Yifan (Evan) Peng (2026). “Boundary-aware Neural Model Reduction for PDEs”. In: *ACM SIGGRAPH Conference Papers 2026*. DOI: 10.1145/3799902.3811153.
- Zixuan Li[†], **Pengfei Shen**[†], Hanxiao Sun, Zibo Zhang, Yu Guo, Ligang Liu, Ling-Qi Yan, Steve Marschner, Milos Hasan, and Beibei Wang (2026). “Fiber-level Woven Fabric Capture from a Single Microscopic Image”. In: *ACM Transactions on Graphics (TOG)* 45.4.
- Zhenyang Li[†], Xiaoyang Bai[†], Tongchen Zhang, **Pengfei Shen**, Weiwei Xu, and Yifan (Evan) Peng (2025). “Enhanced Velocity Field Modeling for Gaussian Video Reconstruction”. In: *Proceedings of IEEE ISMAR 2025*.
- Pengfei Shen**[†], Feifan Qu[†], Li Liao[†], Ruizhen Hu, and Yifan (Evan) Peng (2025). “S³ Imagery: Specular Shading from Scratch-Anisotropy”. In: *Proceedings of ACM SIGGRAPH Asia 2025 (Technical Papers)*.
- Zhenyang Li[†], Xiaoyang Bai[†], Jinfan Lu, **Pengfei Shen**, Edmund Y. Lam, and Yifan (Evan) Peng (2025). “EventTracer: Fast Path Tracing-based Event Stream Rendering”. In: *arXiv preprint arXiv:2508.18071*.
- Feifan Qu, **Pengfei Shen**, Li Liao, and Yifan (Evan) Peng (2025). “Hands-on Light Field Display via Engineering Scalable Optics, Algorithms, and Mounts”. In: *Proceedings of ACM SIGGRAPH Asia 2025 (XR)*.
- Xuchen Wei, Yuchi Huo, **Pengfei Shen**, Yifan Peng, Hujun Bao, and Rui Wang (2025). “Regulation-Aware Freeform Headlamp Reflector Design with Differentiable Ray Tracing”. In: *Optics Express*.
- Ruizeng Li, Xinyang Liu, Runze Wang, **Pengfei Shen**, Ligang Liu, and Beibei Wang (2025). “Bidirectional Plateau-Border Scattering Distribution Function for Realistic and Efficient Foam Rendering”. In: *Proceedings of EGSR 2025*.
- Pengfei Shen**[†], Ruizeng Li[†], Beibei Wang, and Ligang Liu (2023). “Scratch-based Reflection Art via Differentiable Rendering”. In: *ACM Transactions on Graphics (Proceedings of SIGGRAPH 2023)* 42.4, pp. 1–12.
- Tao Zhuang, **Pengfei Shen**, Beibei Wang, and Ligang Liu (2021). “Real-time Denoising Using BRDF Pre-integration Factorization”. In: *Computer Graphics Forum* 40.7, pp. 173–180. DOI: <https://doi.org/10.1111/cgf.14411>.

[†]Equal contribution.

Experience

- 2023 **Technology Researcher Intern**, Tencent Games, Shenzhen, China
Worked on a differentiable renderer for 3D modelling tasks.

- 2023 **Research Assistant**, Nankai University, Tianjin, China
 Participated in a woven fabric reconstruction project; wrote a path tracer rendering billions of fibers at interactive framerates. Collaborated with researchers from UCSB, Adobe, and Cornell.
- 2022 **Graphics Engineer Intern**, MiHoYo Co. Ltd., Shanghai, China
 Implemented *Tessellation-Free Displacement Mapping for Ray Tracing* into Unreal Engine.

Teaching

- 2024 **Teaching Assistant**, Computer Graphics, University of Science and Technology of China
 Lecturer: Prof. Ligang Liu. Main contributor to the course code skeleton.
- 2022 **Teaching Assistant**, Computational Imaging (GAMES 204), Online
 Lecturer: Prof. Qilin Sun & Prof. Evan Peng.
- 2020–21 **Teaching Assistant**, Physics Experiments, USTC
 Designed the open-ended experiment problems.

Talks

- 2023 *Scratch-based Reflection Art via Differentiable Rendering*, GAMES Seminar, Online
- 2020 *Real-time Denoising Using BRDF Pre-integration Factorization*, Pacific Graphics 20+21, Online

Awards

- 2024 USTC Outstanding Teaching Assistant
- 2023 Chinese National Scholarship (top 5%)
- 2023 First Place, CSIG Image and Graphics Challenge — AMD Fluid Simulation Track
- 2019 First Prize, China Undergraduate Physics Tournament

Skills

- Programming C++, CUDA, Python, Vulkan, Unreal Engine, OpenUSD
- Modelling Blender, Houdini
- Languages Chinese (Native), English (Fluent)